Decoding an encoded path in the circle code returns the original path

The Univalent Foundations Program

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Editorial note. The counted statement is the author lem environment at source lines 576-579 and its proof environment opens at line 581 after a blank line; the pairing is by adjacency and the retained proof range 581-587 is the whole environment.

Editorial note. The retained counter values (chapter 8, section 1, theorem offset 6) make the counted lemma print as Lemma 8.1.7, matching the original book numbering read from the label-number dump of the pinned snapshot (newlabel lem:s1-decode-encode = 8.1.7).

Editorial note. No context span and no external reference are needed: the retained statement and proof use no label or citation from outside their own ranges, only the definitions of the two maps.

Editorial note. The retained preamble is the book own macro library and layout adapter as stored for this chapter; no package outside it is requested.

Editorial scope. One original result of the pinned HoTT Book source tree, counted once: the decode-encode identity for the circle code (author label lem:s1-decode-encode, book Lemma 8.1.7).

Allowed standard prerequisites (editorial).

- Circle code and its two maps. The code family assigns types to the points of the circle, encode transports the canonical element along a path, and decode reads an element of the code as a path to the base point.. They are the two maps composed in the statement.
- Path induction. To prove a statement about an arbitrary path from a fixed point it suffices to prove it for the reflexivity path.. It reduces the claim to the base point.
- Computation of the two maps at the base point. Encoding the reflexivity path at the base point is the integer zero, and decoding zero at the base point is the reflexivity path.. They are the two judgemental computations that close the proof.

Lemma 8.1.7. *For all* $x : \mathbb{S}^1$ *and* $p : \mathsf{base} = x$, $\mathsf{decode}_x(\mathsf{encode}_x(p)) = p$.

Proof. By path induction, it suffices to show that $\mathsf{decode}_{\mathsf{base}}(\mathsf{encode}_{\mathsf{base}}(\mathsf{refl}_{\mathsf{base}})) = \mathsf{refl}_{\mathsf{base}}$. But $\mathsf{encode}_{\mathsf{base}}(\mathsf{refl}_{\mathsf{base}}) \equiv \mathsf{transport}^{\mathsf{code}}(\mathsf{refl}_{\mathsf{base}}, 0) \equiv 0$, and $\mathsf{decode}_{\mathsf{base}}(0) \equiv \mathsf{loop}^0 \equiv \mathsf{refl}_{\mathsf{base}}$. □